

CAPSTONE DUAL-BRAIN HETEROGENEOUS SYSTEM ARCHITECTURE

Arduino Uno Q / STM32 NPU: Cortex-A MPU Deliberation ↔ Shared TCM SRAM ↔ Real-Time Cortex-M MCU Reflex Enforcer

1. HOST PROCESSOR (Linux MPU / NPU)

High-Throughput Deliberation

Cadence: 20–50 Hz · Non-Real-Time

- MIPI CSI-2 Camera DMA Ingestion
- Spatial Tokenizer (DINOv2 / MobileNet)
- SE(3) Dynamic Kinematic Frame Tree
- VLM Semantic Intent Grounding
- Diffusion / ACT Action Chunks ($H=16$)
- Asynchronous State Re-Anchoring

STOCHASTIC / UNTRUSTED SUBSTRATE

Subject to page faults, GC stalls, and tail spikes

SHARED MEMORY (TCM / SRAM)

Lock-Free Inter-Core Mailbox & Seqlock

Candidate Action Buffer p_t

$H=16$ Chunk Poses + Velocities (128 bytes)

Expiring Intent Lease t_expire

Monotonic Timestamp Dead-Man Switch (8 bytes)

World Model State $\hat{z}_t$

Fused SE(3) Bounding Primitives (64 bytes)

Hardware Heartbeat Counter

20 Hz Monotonic Ping (4 bytes)

Proprioceptive Telemetry

1 kHz Odometry & Motor Currents (32 bytes)

2. REFLEX ENFORCER (Real-Time MCU)

Deterministic Safety Enforcer

Cadence: 1000 Hz · Hard Real-Time (Jitter < 5 μ s)

- 1000 Hz FreeRTOS Strict Priority Task
- Active-Set CBF QP Solver ($h(x) \geq 0$)
- Dynamic Stopping Clearance Veto ($d_stop \leq d_gap$)
- Watchdog Heartbeat Monitor (50 ms Timeout)
- Bumpless C² Takeover Smoothing $S(\alpha)$
- PWM Gate Driver Current Loop Control

CERTIFIED SAFETY GUARANTEE

Zero dynamic malloc · Strict forward invariance

ACTUATORS & PHYSICAL ENVIRONMENT ($W_t \rightarrow W_{t+1}$)

BLDC Motors · Inverter Bridges · Contact Friction · Mechanical Inertia · Thermal Dissipation